

E033

Azure Solutions Architecture

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A retailer's cloud team of 40 engineers works across three time zones. An outside contractor needs to start, stop, and restart virtual machines in one resource group, rg-web-prod, for a six-week project. The contractor's firm is ISO 27001 certified. Which role assignment follows least privilege?

- A. Contributor at the subscription scope so the contractor is never blocked.
- B. Owner on rg-web-prod.
- C. Virtual Machine Contributor (or a custom VM-operator role) scoped to rg-web-prod.
- D. Reader on rg-web-prod, since the firm's certification already covers risk.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

A user has Reader at the subscription scope and Contributor on one resource group in that subscription. What can the user do in that resource group?

- A. Only read, because the more restrictive role always wins.
- B. Change resources anywhere in the subscription.
- C. Only read, because the higher-scope role wins.
- D. Create and change resources, because the permissions add together.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

Compliance says every resource must be deployed only in East US or West US. The company has 12 subscriptions under one management group, and its security team just finished a Zero Trust workshop. What is the best way to enforce the rule?

- A. Add a Conditional Access policy requiring MFA for all deployers.
- B. Create a custom RBAC role that only allows deployments in those two regions.
- C. Assign an Azure Policy with an allowed-locations rule at the management group.
- D. Put a delete lock on each subscription.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

An order database is backed up every 4 hours. The business says it can lose at most 15 minutes of orders and can be offline for up to 2 hours. The database runs on premium SSD storage and handles about 3,000 transactions a minute. Which requirement does the current backup plan fail?

- A. Throughput, because premium SSD cannot keep up with 3,000 transactions a minute.
- B. The RTO, because 4 hours between backups is longer than 2 hours.
- C. Neither, because a restore can finish within 2 hours.
- D. The RPO, because up to 4 hours of orders could be lost against a 15-minute limit.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

After an outage, service came back 3 hours after it failed, and the newest usable copy of the data was 20 minutes older than the failure. Which objectives do these numbers measure against?

- A. 3 hours against RPO; 20 minutes against RTO.
- B. 3 hours against RTO; 20 minutes against RPO.
- C. Both against RTO.
- D. Both against the monthly uptime SLA only.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A web app must keep running if one datacenter in its region fails, with the least added cost and latency. It serves about 50,000 users, mostly in one country, and marketing plans a big launch next quarter. What should the architect use?

- A. Scale up to one larger VM size.
- B. Spread VM instances across availability zones in the region.
- C. Run a second copy in a paired region.
- D. Place the VMs in an availability set.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

Backup files must survive even if an entire Azure region goes down. The storage account holds about 8 TB, and the finance team opens the files about once a month. Which redundancy setting meets the requirement?

- A. Locally redundant storage (LRS).
- B. Zone-redundant storage (ZRS).
- C. Geo-redundant storage (GRS).
- D. The Archive access tier.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8. [TRANSFER — reversed stem / different variable]

A storage account uses zone-redundant storage (ZRS). Which failure does ZRS NOT protect against?

- A. An outage of the entire region.
- B. A single failed disk drive.
- C. The loss of one availability zone.
- D. A power failure in one server rack.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

The operations team wants a page within about a minute when average CPU on a VM scale set stays above 85% for 5 minutes. They also keep 90 days of application logs for auditors, and an intern built their current dashboard. What should they set up?

- A. A metric alert rule on Percentage CPU with an action group that pages the team.
- B. A log search alert that queries the application logs once a day.
- C. An activity log alert on the scale set.
- D. Raise log retention to 365 days.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A function app reads secrets from Key Vault using a client secret stored in its config file. The file lives in a private Git repo with branch protection and two required reviewers. What change best removes the credential risk?

- A.** Add a third required reviewer to the repo.
- B.** Turn on a managed identity for the function app and grant it a Key Vault secrets-reader role.
- C.** Give the function app the Key Vault Contributor role.
- D.** Move the client secret into an app setting environment variable.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

Checkout runs on a web tier with a 99.95% SLA that must call a database with a 99.99% SLA; both are required. A CDN with its own SLA serves product images but is not needed for checkout. The team that built it has six developers. What is the composite SLA for checkout?

- A.** About 99.94%
- B.** 99.95%
- C.** About 99.99999%
- D.** 99.99%

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12. [TRANSFER — reversed stem / different variable]

Two identical VMs, each 99% available, sit behind a load balancer, and either one can serve all traffic. Ignoring the load balancer, what is their combined availability?

- A.** 98.01%
- B.** 100%, because there is no single point of failure.
- C.** 99%
- D.** 99.99%

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer C Concept 1

A retailer's cloud team ~~of 40 engineers works across three time zones.~~ An outside contractor needs to start, stop, and restart virtual machines in one resource group, rg-web-prod, for a six-week project. ~~The contractor's firm is ISO 27001 certified.~~ Which role assignment follows least privilege?

Target word: role and scope for one narrow task **Decides:** The task is VM operations in one resource group only.

A	Adjacent category	A role that works, but the scope covers every resource group in the subscription.
B	Dropped element	Right scope but drops the narrow action set; Owner can also grant access to others.
C	Correct	Narrow role, lowest scope that covers the work.
D	Camouflage	The certification does not decide access; Reader cannot restart VMs.

2. Answer D Concept 1 · transfer

A user has Reader at the subscription scope and Contributor on one resource group in that subscription. What can the user do in that resource group?

Target word: effective permission when roles overlap **Decides:** Role assignments are additive and inherit downward.

A	Confusion set	That describes a deny assignment, not two allow roles.
B	Absolute language	Contributor applies only to the resource group where it was assigned.
C	Inverted operation	Inheritance adds permissions downward; the parent scope does not override a child grant.
D	Correct	Effective access is the union of Reader and Contributor in that group.

3. Answer C Concept 2

Compliance says every resource must be deployed only in East US or West US. The company has 12 subscriptions under one management group, ~~and its security team just finished a Zero Trust workshop.~~ What is the best way to enforce the rule?

Target word: enforce an allowed-locations rule **Decides:** The rule is about resource properties, across many subscriptions.

A	Camouflage	Answers the Zero Trust detail; sign-in controls do not govern resource location.
B	Confusion set	RBAC decides who can act, not which regions a resource may use.
C	Correct	Policy governs resource settings and inherits to all 12 subscriptions.
D	Adjacent category	Locks prevent deletion or changes; they do not restrict region.

4. Answer D Concept 3

An order database is backed up every 4 hours. The business says it can lose at most 15 minutes of orders and can be offline for up to 2 hours. ~~The database runs on premium SSD storage and handles about 3,000 transactions a minute.~~ Which requirement does the current backup plan fail?

Target word: which objective the backup interval breaks **Decides:** A 4-hour backup interval can lose up to 4 hours of data.

A	Camouflage	Storage performance does not decide recovery objectives.
B	Confusion set	RTO is time to restore service, not time between backups.
C	Dropped element	Checks only RTO and ignores the data-loss limit.
D	Correct	Backup frequency sets the worst-case data loss, which is RPO.

5. Answer B Concept 3 · transfer

After an outage, service came back 3 hours after it failed, and the newest usable copy of the data was 20 minutes older than the failure. Which objectives do these numbers measure against?

Target word: map measured numbers back to RTO and RPO **Decides:** Downtime compares to RTO; data age compares to RPO.

A	Inverted operation	Swaps the two objectives.
B	Correct	Time to recover is RTO; how much data was lost is RPO.
C	Confusion set	Data age is a data-loss measure, not a downtime measure.
D	Adjacent category	An SLA percentage is a different measure from recovery objectives.

6. Answer B Concept 4

A web app must keep running if one datacenter in its region fails, with the least added cost and latency. ~~It serves about 50,000 users, mostly in one country, and marketing plans a big launch next quarter.~~ What should the architect use?

Target word: the right fault-domain level **Decides:** The failure to survive is one datacenter within a region.

A	Dropped element	A single instance still lives in one datacenter.
B	Correct	Zones are separate datacenters in one region, with low latency between them.
C	Adjacent category	Covers a regional outage but adds cost and latency the requirement does not need.
D	Confusion set	Availability sets spread racks and hosts inside one datacenter.

7. Answer C Concept 5

Backup files must survive even if an entire Azure region goes down. ~~The storage account holds about 8 TB, and the finance team opens the files about once a month.~~ Which redundancy setting meets the requirement?

Target word: redundancy that survives a regional loss **Decides:** Only geo-redundant options copy data to a second region.

A	Dropped element	All copies are in one datacenter.
B	Adjacent category	Survives a zone loss, but all copies stay in one region.
C	Correct	GRS replicates to the paired secondary region.
D	Camouflage	Answers the once-a-month access fact; a tier sets cost, not redundancy.

8. Answer A Concept 5 • transfer

A storage account uses zone-redundant storage (ZRS). Which failure does ZRS NOT protect against?

Target word: the limit of ZRS **Decides:** ZRS copies stay inside one region.

A	Correct	ZRS has no copy outside the region.
B	Inverted operation	ZRS does protect against this.
C	Confusion set	Surviving a zone loss is exactly what ZRS adds over LRS.
D	Adjacent category	A rack failure is smaller than a zone failure and is covered.

9. Answer A Concept 6

The operations team wants a page within about a minute when average CPU on a VM scale set stays above 85% for 5 minutes. ~~They also keep 90 days of application logs for auditors, and an intern built their current dashboard.~~ What should they set up?

Target word: fast numeric threshold alert **Decides:** CPU is a platform metric; the alert must be near real time.

A	Correct	Metric alerts evaluate near-real-time numeric data.
B	Adjacent category	Log alerts run on a schedule over records; daily is far too slow.
C	Confusion set	The activity log records management operations, not CPU load.
D	Camouflage	Retention answers the auditor detail, not alerting.

10. Answer B Concept 7

A function app reads secrets from Key Vault using a client secret stored in its config file. The file lives in a private Git repo ~~with branch protection and two required reviewers.~~ What change best removes the credential risk?

Target word: remove the stored credential **Decides:** Managed identity lets the app authenticate with no secret to store.

A	Camouflage	Tightens the repo detail but leaves the secret in place.
B	Correct	No secret exists to leak or rotate.
C	Confusion set	That is a management-plane role; it does not grant reading secret values.
D	Dropped element	The secret still exists and still must be protected and rotated.

11. Answer A Concept 8

Checkout runs on a web tier with a 99.95% SLA that must call a database with a 99.99% SLA; both are required. ~~A CDN with its own SLA serves product images but is not needed for checkout. The team that built it has six developers.~~ What is the composite SLA for checkout?

Target word: serial composite availability **Decides:** Two required components in series multiply.

A	Correct	$0.9995 \times 0.9999 = 0.99940$.
B	Dropped element	Uses only the weaker component and drops the database.
C	Inverted operation	Treats the two as redundant copies (parallel) when both are required.
D	Wrong stem number	The database SLA alone.

12. Answer D Concept 8 · transfer

Two identical VMs, each 99% available, sit behind a load balancer, and either one can serve all traffic. Ignoring the load balancer, what is their combined availability?

Target word: parallel composite availability **Decides:** The service is down only if both fail: 0.01×0.01 .

A	Inverted operation	Multiplies availabilities as if both were required.
B	Absolute language	Both can still fail at once.
C	Dropped element	Ignores the second VM.
D	Correct	$1 - (0.01 \times 0.01) = 0.9999$.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	7	
Wrong stem number	A figure copied from the prompt that is not the answer.	1	
Adjacent category	The right idea in the wrong bucket.	7	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	5	
Absolute language	Always, never, guarantee — where judgment applies.	2	
Confusion set	Every option is real; the test is exact role-matching.	8	
Camouflage	Answers a stem fact that was never the question.	6	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	RBAC least privilege and scope inheritance	Grant the narrowest built-in or custom role at the lowest scope that covers the task; role assignments inherit downward and are additive.	Assigning Owner or Contributor at subscription scope “so nobody gets blocked.”	One person, one task, one resource group.
2	Azure Policy vs. RBAC	RBAC controls who can act; Azure Policy controls what resources and settings are allowed, regardless of who deploys them.	Trying to enforce allowed regions or SKUs with a custom role.	“All resources must...” (a rule about the resource, not the person).
3	RPO vs. RTO	RPO is the most data (measured in time) you can afford to lose; RTO is the longest you can afford to be down.	Comparing backup frequency to RTO instead of RPO.	“Lose at most...” vs. “back online within...”.
4	Availability sets vs. zones vs. regions	Availability sets protect against rack and host failures; availability zones against a datacenter failure; a second region against a regional outage.	Using an availability set to survive a datacenter loss.	The size of the failure the design must survive.
5	Storage redundancy options	LRS keeps copies in one datacenter; ZRS across zones in one region; GRS/GZRS also copy to a paired secondary region.	Assuming ZRS survives a full regional outage.	“If the region goes down” vs. “if a datacenter goes down.”
6	Metrics vs. logs vs. activity log in Azure Monitor	Platform metrics are near-real-time numbers for fast threshold alerts; logs are queried records for analysis; the activity log records control-plane operations.	Building a fast CPU alert on a scheduled log query or the activity log.	A numeric threshold and a need to alert within minutes.
7	Managed identities over stored secrets	Give Azure resources a managed identity and grant it a data-plane role so no credential lives in code or config.	Moving the secret to a “safer” place instead of removing it.	An app authenticating to another Azure service with a stored secret.
8	Composite SLA	Required components in series multiply their availabilities (result is lower than any one); redundant parallel copies combine as 1 – product of failure rates.	Taking the lowest or the average SLA as the composite.	Two or more SLAs and the words “depends on” or “either can serve.”

Course scope (public description): Azure Solution Architecture explores the design and implementation of scalable, secure, and efficient solutions within the Azure platform. This course examines key principles of identity management, governance, monitoring, business continuity, data storage, and infrastructure management to support enterprise cloud environments. Emphasis is placed on aligning architectural decisions with business requirements, optimizing performance, and ensuring security and compliance. Cloud Platform Solutions is a prerequisite for this course.